Beyond the Headline: Calibrated News Article Classifier

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***Abstract*—News topic classifiers usually see only a headline, since that is all an RSS feed carries. I scraped the full article bodies and measured whether the extra text pays. On 8,001 human-labelled articles across 13 topics, the body raises macro-F1 from 0.712 to 0.771, at McNemar $p = 7.5 \times 10^{-6}$. The taught families were measured first: logistic regression 0.752, random forest 0.730, and XGBoost 0.735 at five times the fit cost, while a majority vote of ten models gained 0.001. A linear support vector machine over word and character n-grams reached 0.771 and was kept. Once calibrated it declines 19.3% of test articles and raises accuracy on the rest from 77.7% to 83.4%. Test macro-F1 is 0.751 [0.719, 0.780].**

***Index Terms*—text classification, macro-F1, class imbalance, probability calibration, classification with rejection, support vector machine**

I. Introduction and Problem

A reader who follows *The Hindu*, *The Indian Express* and *Deccan Herald* sees the same PTI wire story three times, filed under a different section name in each paper. Merging those feeds needs every article re-filed under one shared set of 13 topics.

An RSS item carries only a headline, so that is the usual input. A collector that follows the link also has the body, about 100 times more text. More text need not help: bodies carry house style, navigation furniture and author biographies, which a classifier can learn instead of the topic. Whether the body pays is the main question here. The classes are imbalanced at about 6.7 : 1, from 213 validation articles for politics down to 29 for education, so macro-F1 is the primary metric.

II. Related Work

The 13 labels are a flattened subset of IPTC Media Topics [1]. Linear classifiers over term-weighted features have been the reference method for topic classification since Joachims [2]. Zadrozny and Elkan [3] gave the isotonic method used here, Guo *et al.* [4] the calibration error that scores it, and Chow [5] the rejection option. Dietterich [6] recommends a paired test, and Efron and Tibshirani [7] the resampling used for the intervals.

III. Data and Preparation

A Go service polls 97 RSS and Atom feeds into MongoDB and scrapes the linked bodies. The corpus holds 14,189 articles from 40 publishers, 92.6% with a body, and 8,001 carry a single human label over 13 topics. Per-publisher line counting found 355 pieces of furniture across 47 sources; *"Story continues below this ad"* appears in 75% of one publisher's bodies. Republished copies of one story are then grouped together, which I call near-duplicate grouping, since a PTI report carried by three papers is one story. It reaches precision 0.86 at recall 0.81 on 43 hand-judged pairs, and stops a syndicated copy of a training article from leaking into validation. The splits hold 5,487 training, 1,120 validation and 1,159 test articles.

IV. Method

Features are TF-IDF vectors, which score a term highly when it is frequent in one article and rare across the corpus, so *repo rate* counts for more than *said*. The base features are word n-grams, runs of one or two consecutive words.

I started with the taught models; logistic regression reached 0.752 macro-F1 on the body variant. K-nearest neighbours was excluded by reasoning rather than by a run: it is a lazy learner comparing each query against all 5,487 training articles in roughly 30,000 dimensions, where the curse of dimensionality leaves distances nearly equal for every pair. Tree ensembles need dense input, so the features were reduced to 256 components; random forest reached 0.730, extra trees 0.684 and XGBoost 0.735 at five times the fit time.

The taught families plateaued near 0.75, so I read further and tried a support vector machine, which places the boundary so that the gap to the nearest training points of each class is as wide as possible. That gap is the margin, and the score $w \cdot x + b$ is the same linear quantity logistic regression pushes through the sigmoid. The plain SVM only ties logistic regression, 0.753 against 0.752; the gain came from character n-grams, runs of three to five consecutive characters, which lifted it to 0.771 because *bowled*, *bowler* and *bowling* then share evidence. The final model is a `LinearSVC` ($C = 1$, `class_weight="balanced"`) over word 1–2 and `char_wb` 3–5 grams, reading the title plus the first 4,000 body characters.

An SVM margin has no fixed scale and is not comparable across the 13 classes, so it must never be shown as a confidence. Probability calibration adjusts scores so that a stated confidence of 0.8 is right about 80% of the time. I use isotonic regression [3], a non-decreasing step function from raw score to observed accuracy, over five grouped folds of train, and abstain below one global cut of 0.584.

TABLE I
VALIDATION MACRO-F1: THE BODY A/B TEST, THEN PUBLISHER HOLDOUTS.

Candidate	t+summary	t+body	Hindu	Guardian
ComplementNB	**0.635**	0.594	0.589	0.563
logistic regression	0.698	0.752	0.747	0.736
linear SVM, words	0.696	0.753	0.750	0.745
word+char SVM	0.712	**0.771**	0.769	0.724
XGBoost, 256 comp.	—	0.735	0.721	0.679
random forest	—	0.730	0.705	0.695
MiniLM embeddings	—	0.710	0.718	0.693

V. Evaluation Protocol

With 6.7 : 1 imbalance, accuracy alone would hide failure on the small classes, so the metrics come from the 13-class confusion matrix.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

$$P_i = \frac{TP_i}{TP_i + FP_i}, \qquad R_i = \frac{TP_i}{TP_i + FN_i} \tag{2}$$

$$F1_i = \frac{2\,P_i\,R_i}{P_i + R_i} \tag{3}$$

$$\text{macro-F1} = \frac{1}{13}\sum_{i=1}^{13} F1_i \tag{4}$$

Macro-F1 in (4) weights every class equally, so education with 29 articles counts as much as politics with 213. Weighted averaging lets the large classes dominate and micro averaging reduces to accuracy, the trap this dataset sets. The split is grouped and temporal: the cuts follow collection time and no near-duplicate group crosses one. Intervals are bootstrapped over story groups rather than articles [7], since copies of one wire story are not independent samples. Comparisons use McNemar's test, a paired test over only the articles where two models disagree [6]. *The Hindu* and *The Guardian* are held out and rescored on every run with the model refitted without them. The test split was opened once, after the model was frozen.

ROC and AUC are not reported. AUC scores only how well a model ranks articles and does not change if every probability shifts by a constant. This system acts on the probability itself, comparing it against a cut of 0.584, so a ranking score would miss the property it depends on. I report expected calibration error instead, the gap between stated confidence and observed accuracy.

VI. Results

Table I answers the main question. The body is worth +0.059 macro-F1 on the final model, at McNemar $p = 7.5 \times 10^{-6}$ with disjoint intervals, and the gain holds on every linear model. ComplementNB, a Naive Bayes variant adapted for imbalanced text, is the one model that gets worse with more text, because its term-independence assumption weakens as documents lengthen. Bodies were expected to carry publisher style and hurt an unseen masthead; instead *The Hindu* rose from 0.682 to 0.769 and *The Guardian* from 0.665 to 0.724.

No other family caught the linear model. Text is close to linearly separable in a very high-dimensional sparse space, and reducing it to 256 dense components to make trees usable discards the detail the linear model uses. MiniLM embeddings, a small neural encoder that maps text to a fixed-length vector, landed at 0.710, because it truncates at 256 tokens and never reaches the body.

TABLE II
THE HELD-OUT TEST SPLIT, OPENED ONCE.

	validation	test
macro-F1	0.769 [0.738, 0.794]	**0.751 [0.719, 0.780]**
accuracy	0.795	0.778
expected calibration error	0.039	**0.021**
coverage / accuracy at cut	0.761 / 0.879	**0.807 / 0.834**

An oracle picking the correct member among the ten candidates would score 0.900 $[0.878, 0.919]$, so the candidates do disagree usefully. Every majority vote still landed on the incumbent: the best member set scored 0.772 against 0.771, at McNemar $p = 1.0000$. Voting pays only when the members' errors are roughly independent, and here they are asymmetric: MiniLM disagrees on 271 validation articles, rescues 70 and breaks 148 that were already right.

Calibration costs almost nothing in accuracy, 0.769 against a raw 0.771, and the reliability curve sits above the diagonal, so the model is under-confident. One global cut was enough; per-class cuts scored 0.879 against 0.891 at matched coverage. Declining the least certain 19.3% of test articles raises accuracy on the rest to 83.4%, and the validation-to-test gap of −0.021 sits inside the ±0.05 guard.

VII. Limitations

society_lifestyle scores F1 0.43 on validation and 0.44 on test, because the class is a grab-bag of community, labour and lifestyle stories; it still reaches 91% precision at a high cut, so abstention protects it. *education* has only 29 validation articles, so its 0.72 is noise. About 18.6% of validation errors sit on class pairs the annotators themselves disagreed on, so macro-F1 has a real ceiling below 1.0. The window is four days, the system is English-only and India-heavy, and the bundle is 148.3 MB.

VIII. Conclusion

Reading the article body instead of the headline and summary is worth +0.059 macro-F1 at $p = 7.5 \times 10^{-6}$. Logistic regression, random forests, boosted trees and embeddings were measured on the same splits and none beat a linear support vector machine over word and character n-grams. On unseen data the model scores 0.751 macro-F1 and files 80.7% of articles at 83.4% accuracy.

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